

Training and Evaluation of Machine Learning Models in Air Quality Forecasting

Abstract—Air quality forecasting is one of the necessities research areas that are being exploited to enforce all tools and processes that we need to manage the problem of pollution in urban areas and to deal with public health and regulation issues.

Therefore, it is necessary to develop techniques that can be easily deployed to perform pollution forecasting tasks in order to predict the concentration level of air pollutants and its trend in both the short-term and long-term; this techniques should be able integrated to ensure the best possible results while working in a cost-effective way.

The aim of this project was to train and evaluate different Machine Learning models to do air quality forecasting using data taken from different places around the world in order to find the best ones in making predictions and understand how the results we obtained could be useful in future works.

I. INTRODUCTION

Air pollution is an increasingly concerning problem that more and more countries are facing nowadays; it is mainly caused by the population growth and its concentration in urban areas, which are characterized by high industrialization, vehicular traffic emissions and energy consumptions.

In addition to real-time pollution detection, short-term and long-term air quality forecasting is gaining importance since it makes easier to take preemptive actions and mitigate the dangerous effect that an high concentration of pollutants could have on population living in urban areas.

High pollution may have multiple causes: human activities are the main contributors to the problem and have different effect depending also on their distribution over time; however, other factors regarding meteorological conditions as wind, rain, humidity and others can also have some influence on the pollution levels.

The major pollutant that is usually considered in air quality analysis is PM_{2.5}, which consists in particles with a diameter of approximately 2.5 micrometers that can easily penetrate into cardiovascular respiratory systems leading to a negative impact on human health.

The aim of this work is focused on using Machine Learning techniques to predict air quality using meteorological and sensor data collected across different places in order to evaluate how well is possible to predict PM_{2.5} levels in the next hours or days. We will show which models resulted to be better suited for air quality forecasting, and how they behave when working on data gathered from different places and different periods.

We will also explain how this techniques could contribute when integrated with other processes to reach better performance in air quality prediction.

II. PROBLEM STATEMENT

A. Objective

The purpose of this work was to train different Machine Learning models in a way that given an input with feature values at a certain timestamp, the models are able to predict the level of PM_{2.5} in the future (i.e. in the next hours or days); predictions were referred to multiple instants in the future in a range that goes from 1 hour to 1 week with respect to the time-step of the features, and were evaluated using different metrics in order to observe which models performed best.

B. Dataset

During this project we worked on data taken from 11 different locations and different periods, organized in 13 different datasets; each dataset belongs to one single location. Datasets differs on number of samples, number and type of features, number of sensors and periods considered for the measurements, which generally have the duration of about some months:

Dataset	Period	features	Samples
Aosta	04/02/2024 - 31/03/2024	21	1342
Badajoz	12/03/2021 - 17/05/2021	23	1576
Bangalore	21/06/2019 - 31/07/2020	11	7441
Calgary	07/12/2018 - 26/04/2019	10	3026
Dehli	24/07/2018 - 03/01/2020	11	7504
Hamirpur	22/03/2020 - 10/01/2021	11	3638
UK sps030	01/07/2020 - 03/07/2021	35	7330
UK pm5003s	01/07/2020 - 03/07/2021	35	7330
Lima Airbeam	15/11/2021 - 09/01/2022	23	1231
Lima IqAir	15/11/2021 - 14/01/2022	28	1458
Reggio	09/05/2024 - 18/12/2024	11	2511
Torino	01/07/2021 - 08/07/2023	14	12563
Trento	11/07/2024 - 03/02/2025	12	2118

TABLE I: Dataset used in this work

Every dataset contains the values of the following parameters:

- Timestamp
- Relative humidity
- Temperature
- Wind speed
- Rain
- Pressure
- Sensor PM_{2.5} measurement
- Station PM_{2.5} measurement

Sensor data are the values of PM_{2.5} detected from the low-cost sensors and were used as a normal features.

Station measurement is the PM_{2.5} level measured by the stations, which are more precise and reliable than sensors and therefore were considered as the target variable.

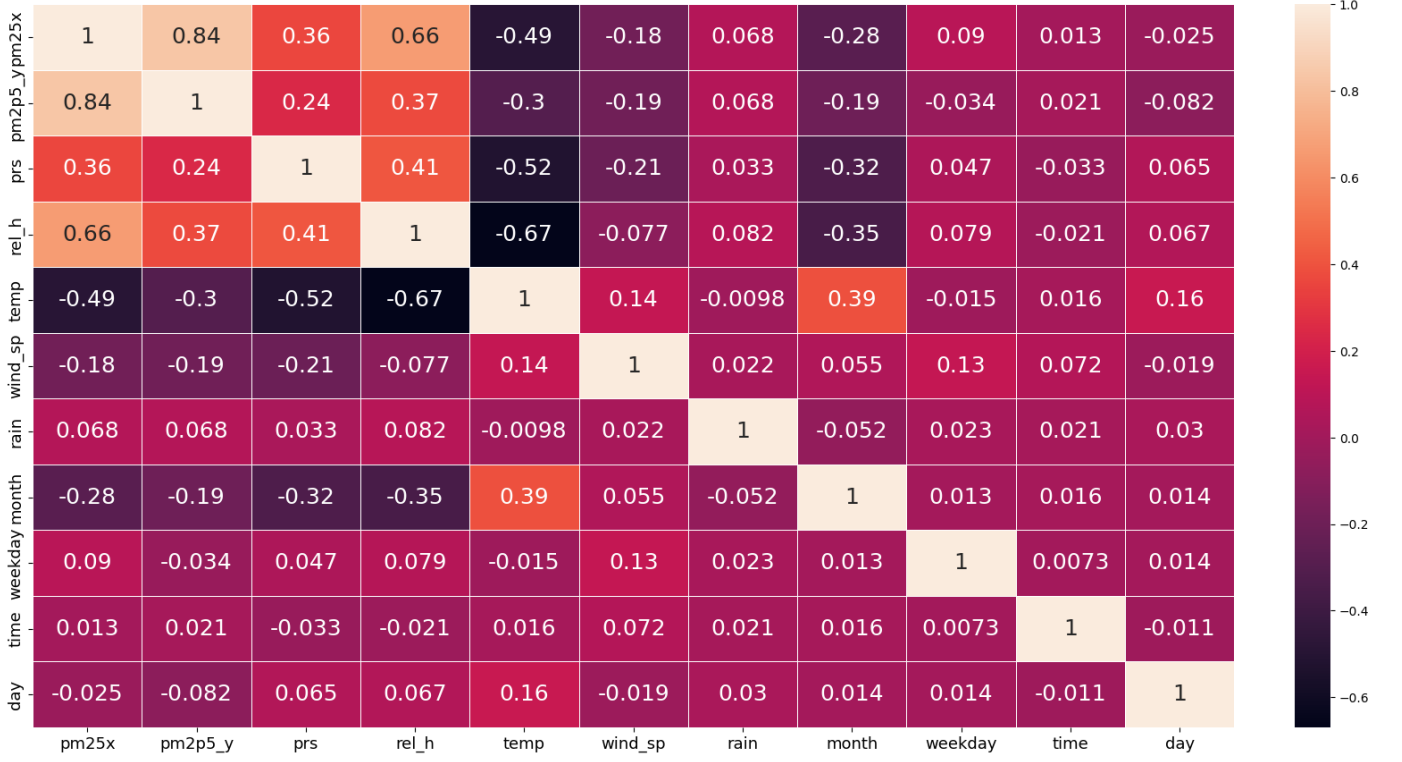


Fig. 1: Correlation matrix for Calgary dataset

Datasets were non structured in the same way, as some of them had others features in addition to those previously presented: measurements of multiple sensors, levels of other air pollutants, such as PM10 or CO2 or others meteorological parameters.

Furthermore, some more features have been added to represent hours, day, week and month, which values were extracted from the timestamp feature and encoded with integers.

In order to understand how each feature influences on the target variable we built a correlation matrix, which is generally used to display the correlation coefficient between pairs of variables.

As expected, sensors data are highly correlated with pm2p5_y; furthermore, we noted that in most of the dataset pressure, temperature wind and relative humidity have a significant effect on target, while among temporal variables only month seems to have a slight impact.

In figure 1 and table II, we show the correlation matrix and coefficients for Calgary dataset: pairs features-target with a value of the coefficient closer to +1 or -1 are more positively or negatively correlated with the target variable.

Features	Correlation
pm2p5_x	0.836710
pressure	0.239064
relative_humidity	0.368142
temperature	-0.301249
wind_speed	-0.193104
rain	0.068213
month	-0.193768
weekday	-0.034302
time	0.021331
day	-0.081649

TABLE II: Correlation coefficients between features and pm2p5_y. Dataset: Calgary

C. Models

Different models were evaluated in order to find the best ones for each dataset, so that the pollution levels forecasting for every location could be performed with the models that best represent the location's characteristics. The following models were used:

- Linear Regression
- Support Vector Regression
- Linear Support Vector Regression
- Nu Support Vector Regression
- Kernel Ridge Regression
- Decision Tree Regression
- Random Forest Regression
- LGBM Regression
- Bayesian Ridge Regression
- Gradient Boost Regressor
- Elastic Net

Every model was evaluated using four metrics: Mean Squared Error, Root Mean Squared Error, Mean Absolute Error and the Coefficient of Determination R2.

III. METHODOLOGY

A. Dataset preparation

All the datasets were modified to make them suited for our task. Firstly, in some datasets there was a lack of data for certain periods, so we did a time interpolation in order to fill the timestamps which initially didn't have any measurement. Then, four more feature representing time, days, weeks and months were added in addition to those already present and were encoded as integers. Therefore each of these new features assume a value within the following range:

- Time = [0, 23]
- Day = [1, 31]
- Week = [0, 6]
- Month = [0, 11]

In our experiments, a normalization of the features was performed by applying three different normalization methods: the Standard Scaler, the Min-Max Scaler and the Maximum Absolute Scaler. However, no metrics improvement was observed during models inference.

B. Grid search and model selection

It was necessary to find the best parameters of each model for every single dataset. In order to do this, we built a set of possible values for all parameters of each model.

In order to do this, we performed a grid search: firstly dataset was split in training set and testing set, then for each model we created one instance of it for all possible combinations of parameters contained in the model's set. Instances were firstly trained and tested using a 5-fold cross-validation to evaluate the goodness of each considered configuration. At the end of the cross-validation phase we selected the parameters set of the instance which obtained the best metrics values.

After the grid search, all models, each with its parameters, were trained and evaluated using the metrics values of the outputs to evaluate and select the best ones for each dataset.

Therefore, at the end of model selection a dataset have a list of models chosen for the forecasting part depending on how good they performed during inference.

The entire process was repeated for every single dataset, so that we could select, for each location, only the models that resulted to be better suited on the location's characteristics. For each dataset, the number of models selected for each dataset varied from 2 to 5, depending on how many models obtained good metrics.

Grid search and model selection were applied to the original datasets without shifting the target values, because during experiments it was noticed that using a lagged version of the datasets didn't have influence on neither the final results of the grid search or the prediction.

C. Prediction

At his point we proceeded with the air quality forecasting task. The goal of this phase was to train our models so that, given in input the feature values at a certain time-step, we get in output the predictions about PM2.5 levels at different time-steps in the future, starting from predictions of PM^{2.5} at 1 hour in the future to predictions at 1 week in the future.

In our experiments we tried the following lags to perform forecasting:

- 1 hour
- 2 hours
- 4 hours
- 8 hours
- 12 hours
- 1 day, 24 hours
- 2 days, 48 hours
- 4 days, 96 hours
- 1 week, 168 hours

For each of these, we created a dataset with lagged features shifting the target variable by the same number n of hours, in order to obtain a modified dataset in which samples had the features values of time t and target value of time $t + n$.

Every modified dataset was then split in training dataset, which have the first 75% of samples, and testing dataset, which had the other 25% of samples.

Then we started the forecasting part: for each lag, models were trained on the training dataset using the station measure (pm2p5_y) as ground truth. Then models were used on the test dataset to predict the PM2.5 value at timestamp $t + n$ using features values of timestep t .

Again, each prediction was evaluated using MSE, MAE, RMSE and R2 to compare the prediction performed on test dataset with the true values measured by station.

IV. RESULTS AND ANALYSIS

A. Datasets

Regarding the impact of dataset's characteristics, the metrics of our outputs across all datasets showed that we need at least 3500-4000 samples to avoid underfitting and get a reliable model in air quality forecasting task, and that datasets with a low number of features (between 10 and 14) led to better

performance with respect to the others, probably because higher number of features cause overfitting.

In general the lack of data for different periods in the same dataset was not a problem: in figure 2 we show an example 1-hour forecasting on Hamirpur dataset, the model is only a bit influenced by the lack of data, which is represented by the straight blue line of the caused by the time interpolation

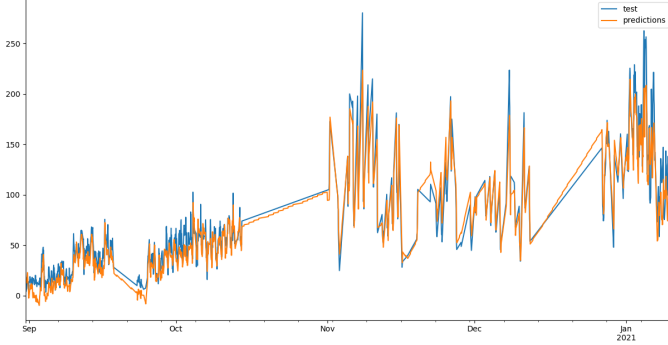


Fig. 2: Dataset: Harmipur, 1-hour forecasting. Model: Bayesian Regression

B. Models

We evaluated the predictions of our models on datasets in order to understand which ones were better suited for the forecasting task. In the table III we show the number of datasets in which every model obtained the best possible performance.

In particular, a model is considered to have good performances if it has R2 metric higher than or equal to 0.50; in cases for one dataset all models had R2 lower than 0.50, we simply accounted the best model for that dataset.

As showed in table III Random Forest Regression and Gradient Boost Regression were the model that demonstrated good performances in predictions in the majority of datasets.

Others models such as LGBM Regression or Kernel Ridge Regression also well performed but only in few cases and obtained bad results in many others, which means that this techniques are more dependent on the dataset characteristics.

Comparing the graphical representation of forecasting with the ground truth, we assessed that, regarding regression models, Random Forest Regressor seems to be the one who better capture the relation between features values and future PM2.5 values.

As we said, Gradient Boost Regression also obtained good performance in doing forecasting but, looking at graphs, in some cases it gives in output predictions of PM2.5 levels that are lagged of a certain amount of time with respect to the ground truth; considerations on this will be discussed later. .

Models	No. Dataset
Random Forest Regression	8
Gradient Boost Regression	8
LGBM Regression	4
Bayesian Regression	3
Kernel Ridge Regression	3
Elastic Net Regression	3
Nu Support Vector Machine	2
Linear Regression	1
Support Vector Machine	1
Linear Support Vector Machine	1

TABLE III: Number of datasets in which every model performed well or generally better than the others. For some datasets, more than one model obtained good results

C. Prediction - metrics

Looking at the metrics used to evaluate our predicted outputs, we observed that good predictions are obtained only for 1, 2 and 4 hours forecasting, with quality of results depending on the dataset. Conversely, for longer time interval predictions, we only obtained an estimate of the evolution that the target variable will have, while the target values predicted were clearly imprecise. In the following table we show an example of metrics values obtained for Delhi dataset using Kernel Ridge Regression:

Metric	1 hour	2 hours	4 hours
R2	0.76	0.71	0.65
MAE	31.93	34.68	37.21
MSE	2790.44	3346.31	4013.39
RMSE	52.82	57.85	63.35

TABLE IV: Metrics for dataset Delhi, Model: Kernel Ridge Regressor (Part 1)

Metric	8 hours	12 hours	1 day
R2	0.63	0.64	0.63
MAE	38.89	38.64	40.14
MSE	4226.82	4153.48	4258.01
RMSE	65.01	64.45	65.25

TABLE V: Metrics for dataset Delhi, Model: Kernel Ridge Regressor (Part 2)

Metric	2 days	4 days	7 days
R2	0.60	0.61	0.58
MAE	42.36	41.96	46.80
MSE	4615.05	4464.97	4886.08
RMSE	67.93	66.82	69.90

TABLE VI: Metrics for dataset Delhi, Model: Kernel Ridge Regressor (Part 3)

D. Prediction - graphical evaluation

We used a graphical representation to compare the ground truth values and evolutions with models prediction. In all reported graphs, the blue line represent the real values of PM2.5 measured by the station; the orange line represent the values predicted by the model.

During our experiments we noticed that in most cases, short-term (usually from 1 to 4 hours forecasting, depending on the dataset) predictions give an acceptable approximation of the real value of PM2.5, although it is rare to obtain an highly precise values forecast.

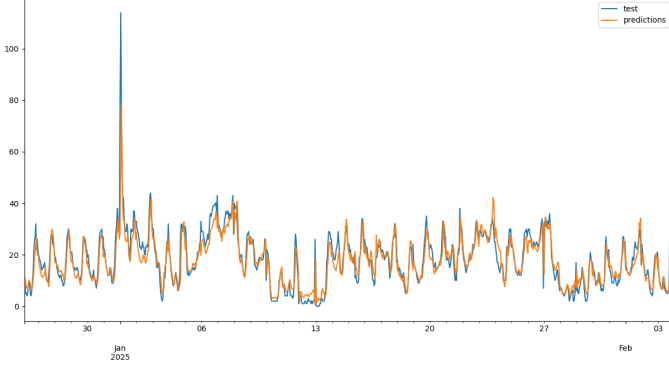


Fig. 3: Dataset: Trento, 1 hour forecasting. Model: Elastic Net Regression

In the long-term forecasting, we noticed that models tends to overestimate low values and underestimate high values, so what we obtain is that predicted PM2.5 levels tend to give a value in the neighborhood of the average level of the considered period.

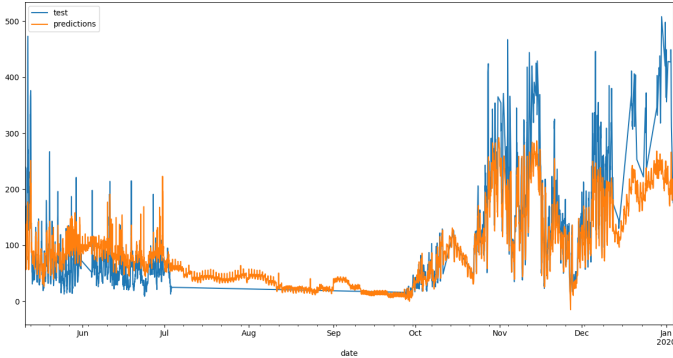


Fig. 4: Dataset: Delhi, 1 hour forecasting. Model: Gradient Boost Regression

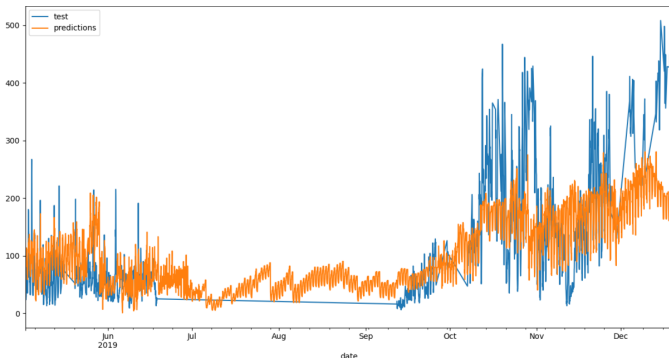


Fig. 5: Dataset: Delhi, 1 week forecasting. Model: Gradient Boos Regression

We also noticed that for some datasets, such as Calgary, models forecasting were characterized by a forward shift of the predicted values with respect to the ground truth of an amount of time equal to or larger than the time difference between the features time-step and the predicted target time-step. In the figure 6 we see an example of this type of error that occurred when we tried to predict PM2.5 at 4 hours in the future: the result is that the model predictions have a lag of some hours forward with respect to the ground truth (blue line).

This means that in some cases the model is not able to well capture the temporal relations between the target and the lagged data, although is still able to predict with a good approximation the PM2.5 level.

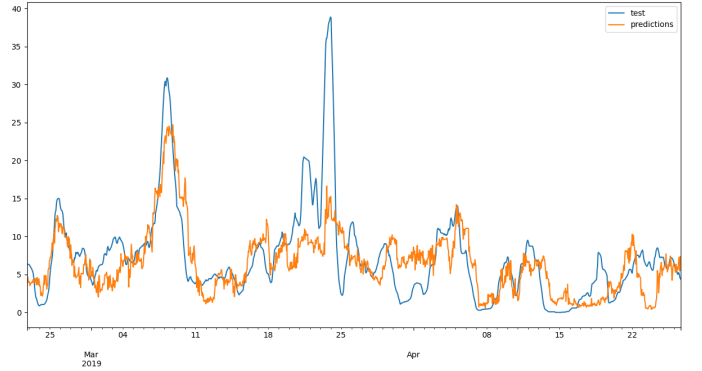


Fig. 6: Dataset: Calgary, 4 hours forecasting. Model: Gradient Boost Regression

To summarize our analysis, we can conclude that, as expected, short-term predictions are much more reliable than long-term ones, and that in general the models often encounter difficulties in predicting the exact value of the target variable.

However, it is worth noting that both in short and long-term forecasting we can predict the evolution that pollutants concentration will follow over time, as showed in figure 5. This suggests that regression techniques could be successfully integrated in a more structured forecasting process where Machine Learning models would have the role of predicting how air quality will change over time, while other components should use this informations and the predictions in order to provide a more precise estimate of the future PM2.5 values.

V. CONCLUSION

In this article we presented an analysis on how Machine Learning models can contribute in air quality forecasting and how their performances are influenced by the characteristics of data coming from different sources and places.

We compared different models acting on multiple datasets and explained how their behavior change from short-term forecasting to long-term forecasting, an gave some indications on which models are better suited for this task.

Having worked with multiple datasets characterized by differences regarding the number of samples, the number and type of features and the lack of data for prolonged periods, we were able to study how this models perform depending on the

nature of each dataset, and we obtained some indications on how dataset should be built to avoid typical Machine Learning problems, as overfitting or underfitting,

Furthermore, another important point to note is that all data used in this project are typical meteorological parameters which are usually easy to obtain and generally publicly available, that were given in input to our models together with low-cost sensor data, so that what we obtain are models that doesn't necessarily need costly measurements taken with more complex instruments.

We also pointed out which are the typical issues that we encountered when doing forecasting regarding the possible presence of shifts or the tendency of ML/DL models to give overestimated or underestimated predictions of the target variable with respect to the the ground truth.

In the end, we hypothesized how the process explained in this paper could be useful when integrated with others forecasting techniques and tools in order to predict with a good estimate the expected pollutants concentration in the next hours or days, and its evolution over time.